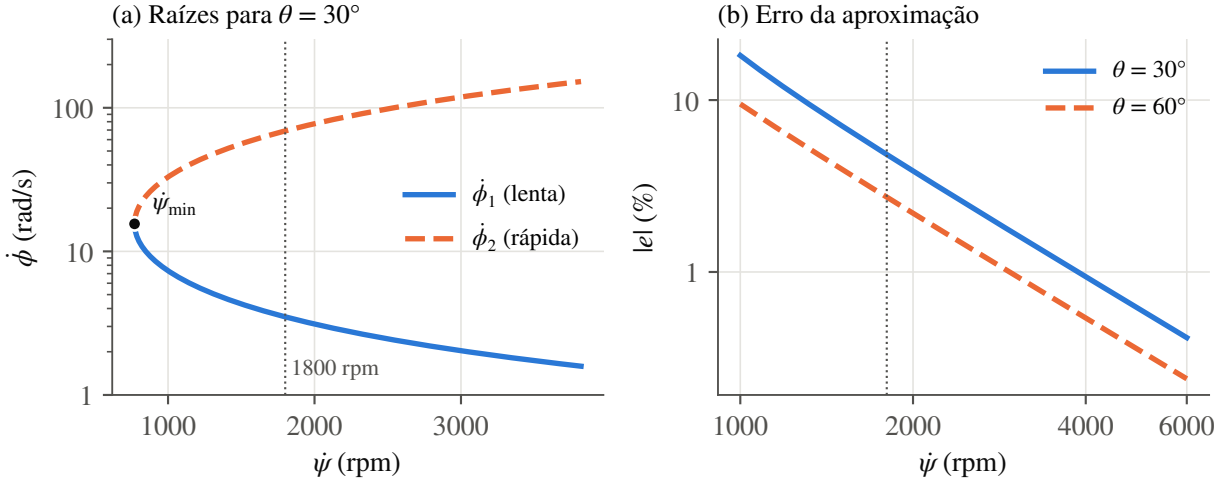
(a) Raízes para $\theta = 30°$
100
10
1
$\dot{\phi}$ (rad/s)
$\dot{\psi}_{\min}$
$\dot{\phi}_1$ (lenta)
$\dot{\phi}_2$ (rápida)
1800 rpm
1000
2000
3000
$\dot{\psi}$ (rpm)
(b) Erro da aproximação
10
1
|e| (%)
$\theta = 30°$
$\theta = 60°$
1000
2000
4000
6000
$\dot{\psi}$ (rpm)